PADHAKU
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CHAPTER 8

Predicting What Comes Next: Exploring Sequences and Progressions

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples; write your own.

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⭐ 8.1 Introduction to Sequences

1 **Describe the pattern in each sequence and predict the next few numbers.**

1, 2, 3, 4, 5, 6, ... (Natural numbers)

1, 3, 5, 7, 9, 11, ... (Odd numbers)

1, 3, 6, 10, 15, 21, ... (Triangular numbers)

1, 4, 9, 16, 25, 36, ... (Square numbers)

✔ ANSWER

- Natural numbers: add 1 each time → **7, 8, 9, ...**
- Odd numbers: add 2 each time → **13, 15, 17, ...**
- Triangular numbers: add 2, 3, 4, 5, 6, then 7, 8, 9 ... → **28, 36, 45, ...**
- Square numbers: 1^2, 2^2, 3^2, ... (add 3, 5, 7, 9, 11, then 13, 15 ...) → **49, 64, 81, ...**

? **Can you think of other finite sequences that you see in your daily life?**

💡 SAMPLE ANSWER

- Days of the week: Monday, Tuesday, ..., Sunday (7 terms).
- Roll numbers in a class: 1, 2, 3, ..., 40.
- Floor numbers of a building: 0, 1, 2, ..., 10.
- Dates of a month: 1, 2, 3, ..., 30.

? **Can you draw the patterns for the next two terms of the triangular numbers (Fig. 8.1)?**

✔ ANSWER

6th triangular number: a triangle of dots with rows 1, 2, 3, 4, 5, 6 → 1 + 2 + 3 + 4 + 5 + 6 = **21 dots**.

7th triangular number: rows 1 to 7 → 21 + 7 = **28 dots**.

? **Fig. 8.2 shows square numbers and odd numbers. Can you explain the relationship?**

✔ ANSWER

Start with 1 dot (1 × 1 square).

Add an L-shaped border of 3 dots → 2 × 2 square (1 + 3 = 4).

Add an L-shaped border of 5 dots → 3 × 3 square (1 + 3 + 5 = 9).

Each new border has 2 more dots than the last, i.e. the next odd number.

So **the sum of the first n odd numbers = n^2**.

Exercise **Consider the sequence 1, 4, 7, 10, 13, ... Predict the next four terms. Derive the first 10 terms of the sequence obtained by adding all the terms up to a given term (1, 1 + 4 = 5, 1 + 4 + 7 = 12, ...).**

✔ ANSWER

The terms go up by 3 each time.

Next four terms: **16, 19, 22, 25**

Running sums:

1, 1 + 4 = 5, 5 + 7 = 12, 12 + 10 = 22, 22 + 13 = 35, 35 + 16 = 51, 51 + 19 = 70, 70 + 22 = 92, 92 + 25 = 117, 117 + 28 = 145

First 10 terms: **1, 5, 12, 22, 35, 51, 70, 92, 117, 145**

Exercise **Can you write t_5, t_6, t_7 and t_8 for the sequence of triangular numbers?**

✔ ANSWER

$t_5 = 1 + 2 + 3 + 4 + 5 = \mathbf{15}$

$t_6 = 15 + 6 = \mathbf{21}$

$t_7 = 21 + 7 = \mathbf{28}$

$t_8 = 28 + 8 = \mathbf{36}$

T&R **Can you think of any other kinds of sequences? List five different types of sequences and discuss their properties.**

💡 SAMPLE ANSWER

- Constant sequence: 5, 5, 5, 5, ... (all terms equal).
- Doubling sequence: 1, 2, 4, 8, 16, ... (each term is 2 times the previous one).
- Alternating sequence: 1, −1, 1, −1, ... (sign changes each time).

- Cube numbers: 1, 8, 27, 64, ... ($t_n = n^3$, grows fast).
- Decreasing sequence: 10, 8, 6, 4, ... (each term 2 less; terms become negative later).

Other examples: prime numbers 2, 3, 5, 7, 11, ... (no simple rule); Virahānka–Fibonacci 1, 2, 3, 5, 8, ...

⭐ 8.2 Explicit Rule for a Sequence

T&R **Why is it useful to have an explicit formula for the nth term of a sequence?**

✔ ANSWER

- We can find any term (20th, 300th, ...) directly by putting the value of n, without knowing the previous terms.
- We can check whether a given number is a term of the sequence, and find its position, by solving t_n = that number.

Exercise **Using the explicit rule $u_n = 2n - 1$, find the 53rd term, the 108th term and the 1170th term of the odd number sequence.**

✔ ANSWER

$u_{53} = 2 \times 53 - 1 =$ **105**

$u_{108} = 2 \times 108 - 1 =$ **215**

$u_{1170} = 2 \times 1170 - 1 =$ **2339**

Example 2 For $s_n = 5n - 2$, write the first 6 terms. What is the 100th term? The 1000th term?

✔ ANSWER

$s_1 = 3$, $s_2 = 8$, $s_3 = 13$, $s_4 = 18$, $s_5 = 23$, $s_6 = 28$

First 6 terms: **3, 8, 13, 18, 23, 28**

$s_{100} = 5 \times 100 - 2 =$ **498**

$s_{1000} = 5 \times 1000 - 2 =$ **4998**

? Can you explain why we need n to be a natural number?

✔ ANSWER

n is the position of a term: 1st, 2nd, 3rd, ...
Positions are counting numbers.

There is no 94.6th position, so if solving gives n = 94.6, **the number is not a term** of the sequence.

T&R Can you find the rule describing the nth term of the sequence of square numbers?

✔ ANSWER

$1 = 1^2$, $4 = 2^2$, $9 = 3^2$, $16 = 4^2$, ...

$t_n = n^2$

? First ten primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. Do you see any pattern? Can you think of a

rule that can predict the next few prime numbers?

💡 SAMPLE ANSWER

The differences are 1, 2, 2, 4, 2, 4, 2, 4, 6 — they do not follow a fixed pattern.

All primes after 2 are odd, but not every odd number is prime (9, 15, 21 are not).

There is no simple rule that gives the next prime; we must check each number. The next primes are 31, 37, 41, 43, 47.

Exercise

Consider $t_n = 3n - 7$.

(i) Find its first, second, third, 12th, 18th and 50th terms.

(ii) Which term of the sequence is 332?

(iii) Is 557 a term of this sequence? Why or why not?

✔ ANSWER

(i) $t_1 = 3 - 7 = \mathbf{-4}$, $t_2 = 6 - 7 = \mathbf{-1}$, $t_3 = 9 - 7 = \mathbf{2}$

$t_{12} = 36 - 7 = \mathbf{29}$, $t_{18} = 54 - 7 = \mathbf{47}$, $t_{50} = 150 - 7 = \mathbf{143}$

(ii) $3n - 7 = 332 \rightarrow 3n = 339 \rightarrow n = 113$

332 is the **113th term**.

(iii) $3n - 7 = 557 \rightarrow 3n = 564 \rightarrow n = 188$

188 is a natural number, so **yes, 557 is the 188th term**.

8.3 Recursive Rule for a Sequence

? For 1, 4, 7, 10, 13, ... verify that $t_n = 3n - 2$.

✔ ANSWER

$t_1 = 3 - 2 = 1$, $t_2 = 6 - 2 = 4$, $t_3 = 9 - 2 = 7$, $t_4 = 12 - 2 = 10$, $t_5 = 15 - 2 = 13$ ✓

The rule gives the sequence.

Example 3 For $u_1 = 1$, $u_n = 2u_{n-1} + 3$ $(n \geq 2)$, the first four terms are 1, 5, 13, 29. Is 133 a term of this sequence?

✔ ANSWER

$u_5 = 2 \times 29 + 3 = 61$

$u_6 = 2 \times 61 + 3 = 125$

$u_7 = 2 \times 125 + 3 = 253$

The terms keep increasing and 133 lies between 125 and 253.

No, 133 is not a term.

? **Virahānka–Fibonacci sequence 1, 2, 3, 5, 8, 13, 21, 34, ... Can you write the next two terms?**

✔ ANSWER

21 + 34 = 55

34 + 55 = 89

Next two terms: **55, 89**

Exercise Set 8.1

1 **Find the first five terms of the sequence in which the nth term is given by (i) $t_n = 3n - 4$, (ii) $t_n = 2 - 5n$, and (iii) $t_n = n^2 - 2n + 3$ for $n \geq 1$.**

✔ ANSWER

(i) $t_n = 3n - 4$:

$t_1 = -1$, $t_2 = 2$, $t_3 = 5$, $t_4 = 8$, $t_5 = 11$ → **−1, 2, 5, 8, 11**

(ii) $t_n = 2 - 5n$:

$t_1 = -3$, $t_2 = -8$, $t_3 = -13$, $t_4 = -18$, $t_5 = -23$ → **−3, −8, −13, −18, −23**

(iii) $t_n = n^2 - 2n + 3$:

$t_1 = 1 - 2 + 3 = 2$, $t_2 = 4 - 4 + 3 = 3$, $t_3 = 9 - 6 + 3 = 6$, $t_4 = 16 - 8 + 3 = 11$, $t_5 = 25 - 10 + 3 = 18$

→ **2, 3, 6, 11, 18**

2 **Find the 10th and 15th terms of the sequence $t_n = 5n - 3$ for $n \geq 1$.**

✔ ANSWER

$t_{10} = 5 \times 10 - 3 =$ **47**

$t_{15} = 5 \times 15 - 3 =$ **72**

3 **Determine whether 97 and 172 are terms of the sequence $t_n = 5n - 3$ for $n \geq 1$.**

✔ ANSWER

$5n - 3 = 97 \rightarrow 5n = 100 \rightarrow n = 20$ (natural number)

97 is a term (20th term).

$5n - 3 = 172 \rightarrow 5n = 175 \rightarrow n = 35$ (natural number)

172 is a term (35th term).

4 **Which term of the sequence $t_n = 5n - 3$ for $n \geq 1$ is 607?**

✔ ANSWER

$5n - 3 = 607 \rightarrow 5n = 610 \rightarrow n = 122$

607 is the **122nd term**.

5 **A sequence is given by the recursive rule $t_1 = -5$, $t_{n+1} = t_n + 3$ for $n \geq 1$. Find the first five terms. Is 52 a term of this sequence? If so, which term is it?**

✔ ANSWER

$t_1 = -5$, $t_2 = -2$, $t_3 = 1$, $t_4 = 4$, $t_5 = 7$

First five terms: **−5, −2, 1, 4, 7**

Each term is 3 more, so $t_n = -5 + (n - 1) \times 3 = 3n - 8$.

$3n - 8 = 52 \rightarrow 3n = 60 \rightarrow n = 20$

Yes, 52 is the 20th term.

6 Let $T_1 = 1$, $T_2 = 2$, $T_3 = 4$, and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$ for $n \geq 4$. Find T_4, T_5, T_6, T_7 and T_8.

✔ ANSWER

$T_4 = 4 + 2 + 1 = \mathbf{7}$

$T_5 = 7 + 4 + 2 = \mathbf{13}$

$T_6 = 13 + 7 + 4 = \mathbf{24}$

$T_7 = 24 + 13 + 7 = \mathbf{44}$

$T_8 = 44 + 24 + 13 = \mathbf{81}$

⭐ 8.4 Arithmetic Progressions

T&R Growing pattern of squares 1, 5, 9, 13 (Fig. 8.3). Predict the number of squares in Stages 5 and 6, in Stages 10, 11 and 12, in Stage 20 and at any stage.

✔ ANSWER

4 squares are added at each stage, so $t_n = 1 + (n - 1) \times 4 = 4n - 3$.

Stage 5: **17**, Stage 6: **21**

Stage 10: **37**, Stage 11: **41**, Stage 12: **45**

Stage 20: $4 \times 20 - 3 = \mathbf{77}$

Stage n: **4n − 3**

T&R Consider all the sequences discussed so far in this chapter. Which ones are arithmetic

progressions and which ones are not? Justify.

✔ ANSWER

APs (same difference between consecutive terms):

- Natural numbers (d = 1), odd numbers (d = 2)
- 1, 4, 7, 10, 13, ... (d = 3); −7, −3, 1, 5, 9, ... (d = 4)
- Sequences from $t_n = 3n - 4$, $2 - 5n$, $5n - 3$, $5n - 2$, $3n - 7$ and $t_1 = -5$, $t_{n+1} = t_n + 3$
- Squares pattern 1, 5, 9, 13, ... (d = 4)

Not APs (difference changes):

- Triangular numbers (differences 2, 3, 4, ...), square numbers (3, 5, 7, ...)
- 6, 12, 24, 48, 96; 1, 1/2, 1/3, 1/4, ...; primes
- 1, 5, 13, 29, ...; 3, 6, 30, 870; Virahānka–Fibonacci; T_n (1, 2, 4, 7, 13, ...); $n^2 - 2n + 3$ (2, 3, 6, 11, ...)

Exercise

Verify that the following are APs and write their nth terms. What do you observe when you plot the ordered pairs?

(i) 2, 5, 8, 11, ...

(ii) −5, −1, 3, 7, ...

✔ ANSWER

(i) 5 − 2 = 8 − 5 = 11 − 8 = 3, so it is an AP with a = 2, d = 3.

$t_n = 2 + (n - 1) \times 3 =$ **$3n - 1$**

(ii) $-1-(-5)=3-(-1)=7-3=4$, so it is an AP with $a=-5$, $d=4$.

$t_n=-5+(n-1)\times 4=$ **$4n-9$**

Plotting (1, 2), (2, 5), (3, 8), (4, 11) and (1, −5), (2, −1), (3, 3), (4, 7): **in each case the points lie on a straight line.**

Exercise **Using $t_n=a+(n-1)\times d$, find the nth term:**

(i) 1/2, 5/2, 9/2, 13/2, ...

(ii) 1.5, 3.5, 5.5, 7.5, ...

✔ ANSWER

(i) $a=1/2$, $d=5/2-1/2=2$

$t_n=1/2+(n-1)\times 2=2n-3/2=$ **$(4n-3)/2$**

(ii) $a=1.5$, $d=2$

$t_n=1.5+(n-1)\times 2=$ **$2n-0.5$**

Exercise **Find recursive rules for the APs in the previous exercises.**

✔ ANSWER

- 2, 5, 8, 11, ...: **$t_1=2$, $t_n=t_{n-1}+3$** for $n\geq 2$
- −5, −1, 3, 7, ...: **$t_1=-5$, $t_n=t_{n-1}+4$** for $n\geq 2$
- 1/2, 5/2, 9/2, ...: **$t_1=1/2$, $t_n=t_{n-1}+2$** for $n\geq 2$
- 1.5, 3.5, 5.5, ...: **$t_1=1.5$, $t_n=t_{n-1}+2$** for $n\geq 2$

Example 5 **A taxi charges a fixed booking fee of ₹200 plus ₹40 per km. If the person**

travels 10 km, what will be the total fare?

✔ ANSWER

Fare for n km = 200 + 40n

For 10 km: 200 + 40 × 10 = 200 + 400

Total fare = **₹600**

⭐ 8.5 Sum of the First n Natural Numbers

T&R **Can the same approach be used to find the sum 1 + 2 + 3 + ... + 100?**

✔ ANSWER

Yes. S = 1 + 2 + ... + 100 and S = 100 + 99 + ... + 1.

Adding: each pair gives 101, and there are 100 pairs.

2S = 100 × 101 = 10100 → S = **5050**

T&R **Can you use the formula $S_n = n(n + 1)/2$ to find S_{20}, S_{50} or S_{1000}?**

✔ ANSWER

S_{20} = 20 × 21 ÷ 2 = **210**

S_{50} = 50 × 51 ÷ 2 = **1275**

S_{1000} = 1000 × 1001 ÷ 2 = **500500**

T&R **The nth triangular number is $t_n = n(n + 1)/2$. Use this to find the 10th, 17th and 80th**

triangular numbers.

✔ ANSWER

$t_{10} = 10 \times 11 \div 2 = \mathbf{55}$

$t_{17} = 17 \times 18 \div 2 = \mathbf{153}$

$t_{80} = 80 \times 81 \div 2 = \mathbf{3240}$

📝 Exercise Set 8.2

1 **Find the 10th and 26th terms of the AP: 3, 8, 13, 18, ...**

✔ ANSWER

$a = 3, d = 5$

$t_{10} = 3 + 9 \times 5 = \mathbf{48}$

$t_{26} = 3 + 25 \times 5 = \mathbf{128}$

2 **Which term of the AP: 21, 18, 15, ... is −81? Also, is 0 a term of this AP? Give reasons.**

✔ ANSWER

$a = 21, d = -3$, so $t_n = 21 + (n - 1)(-3) = 24 - 3n$

$24 - 3n = -81 \rightarrow 3n = 105 \rightarrow n = 35$

−81 is the **35th term**.

$24 - 3n = 0 \rightarrow 3n = 24 \rightarrow n = 8$ (a natural number)

Yes, 0 is the 8th term (21, 18, 15, 12, 9, 6, 3, 0).

3 **Find the nth term of the AP: 11, 8, 5, 2, ... Write the recursive rule for this AP.**

✔ ANSWER

$a = 11, d = -3$

$t_n = 11 + (n - 1)(-3) = \mathbf{14 - 3n}$

Recursive rule: $\mathbf{t_1 = 11, t_n = t_{n-1} - 3}$ for $n \geq 2$

4 **An AP consists of 50 terms in which the 3rd term is 12 and the last term is 106. Find the 29th term.**

✔ ANSWER

$a + 2d = 12$ ... (1)

$a + 49d = 106$ ... (2)

(2) − (1): $47d = 94 \rightarrow d = 2$

$a = 12 - 4 = 8$

$t_{29} = 8 + 28 \times 2 = \mathbf{64}$

5 **How many 2-digit numbers are divisible by 3? What is the sum of all these 2-digit numbers?**

✔ ANSWER

They are 12, 15, 18, ..., 99 (AP with $a = 12, d = 3$).

$99 = 12 + (n - 1) \times 3 \rightarrow 87 = 3(n - 1) \rightarrow n = 30$

30 numbers

Sum = (number of terms) × (first + last) ÷ 2 = 30 × (12 + 99) ÷ 2 = 15 × 111 = **1665**

6 **Harish started work at an annual salary of ₹5,00,000 and received an increment of ₹20,000 each year. After how many years did his income reach ₹7,00,000?**

✔ ANSWER

Increase needed = 7,00,000 − 5,00,000 = ₹2,00,000

Number of increments = 2,00,000 ÷ 20,000 = 10

Salaries form the AP 5,00,000; 5,20,000; ...; 7,00,000 is the 11th term.

After 10 years (in his 11th year) his income reached ₹7,00,000.

7 **A child arranges marbles in rows: 1 in the first row, 2 in the second, 3 in the third, and so on up to 25 rows. How many marbles in all?**

✔ ANSWER

Total = 1 + 2 + ... + 25 = 25 × 26 ÷ 2

= **325 marbles**

⭐ 8.6 Geometric Progressions

T&R **Growing pattern of squares 3, 6, 12, 24 (Fig. 8.6). Predict the number of squares in**

Stages 5 and 6, in Stages 10, 11 and 12, in Stage 20 and at any stage. How is this different from the pattern in Fig. 8.3?

✔ ANSWER

The number doubles each stage: $t_n = 3 \times 2^{n-1}$.

Stage 5: **48**, Stage 6: **96**

Stage 10: 3×2^9 = **1536**, Stage 11: **3072**, Stage 12: **6144**

Stage 20: $3 \times 2^{19} = 3 \times 524288$ = **1572864**

Stage n: $\mathbf{3 \times 2^{n-1}}$

Difference: in Fig. 8.3 the same number (4) is **added** each time (AP); here the number is **multiplied** by 2 each time (GP), so it grows much faster.

Example 6 **Is 1, 2, 4, 8, 16, ... a geometric progression? If so, what is the common ratio?**

✔ ANSWER

2/1 = 4/2 = 8/4 = 16/8 = 2

Yes, a GP with common ratio 2.

Example 7 **Is 1, 3, 9, 27, 81, ... a geometric progression? If so, what is the common ratio?**

✔ ANSWER

3/1 = 9/3 = 27/9 = 81/27 = 3

Yes, a GP with common ratio 3.

Example 8 **Is 1, −1, 1, −1, 1, ... a geometric progression? If so, what is the common ratio?**

✔ ANSWER

−1/1 = 1/(−1) = −1/1 = −1

Yes, a GP with common ratio −1.

Exercise **Check whether the following are GPs and find their nth terms.**

(i) 2, 10, 50, 250, ...

(ii) 4, 8/3, 16/9, 32/27, ...

(iii) 3, −3/2, 3/4, −3/8, ...

✔ ANSWER

(i) 10/2 = 50/10 = 250/50 = 5 → GP, a = 2, r = 5

$t_n = \mathbf{2 \times 5^{n-1}}$

(ii) (8/3) ÷ 4 = 2/3, (16/9) ÷ (8/3) = 2/3, (32/27) ÷ (16/9) = 2/3 → GP, a = 4, r = 2/3

$t_n = \mathbf{4 \times (2/3)^{n-1}}$

(iii) (−3/2) ÷ 3 = −1/2, (3/4) ÷ (−3/2) = −1/2, (−3/8) ÷ (3/4) = −1/2 → GP, a = 3, r = −1/2

$t_n = \mathbf{3 \times (-1/2)^{n-1}}$

Exercise **Find a recursive rule for $t_n = 3 \times 10^{n-1}$ that generates the GP 3, 30, 300, 3000, ...**

✔ ANSWER

Each term is 10 times the previous term.

$t_1 = 3$, $t_n = 10 \times t_{n-1}$ for $n \geq 2$

🤔 Think and Reflect · 8.6.1 Fun with Fractals

(a) **How many black triangles are there in Stages 0 to 3 of the Sierpiński triangle (Fig. 8.7)?**

✔ ANSWER

Each black triangle becomes 3 smaller ones at the next stage.

Stage 0: **1**, Stage 1: **3**, Stage 2: **9**, Stage 3: **27**

(b) **Can you predict the number of black triangles at Stages 4 and 5?**

✔ ANSWER

Stage 4: $27 \times 3 =$ **81**

Stage 5: $81 \times 3 =$ **243**

(c) **Can you find a rule for the number of black triangles at the nth stage?**

✔ ANSWER

$1 = 3^0$, $3 = 3^1$, $9 = 3^2$, $27 = 3^3$, ...

$t_n = 3^n$ (n = stage number)

(d) **If the black area in Stage 0 is 1 square unit, what is the black area in Stages 1, 2 and 3? In Stages 4 and 5? Find a rule for Stage n. What happens to this area as n increases?**

✔ ANSWER

Each stage keeps 3/4 of the black area of the previous stage.

Stage 1: **3/4**, Stage 2: **9/16**, Stage 3: **27/64** square unit

Stage 4: **81/256**, Stage 5: **243/1024** square unit

Rule: $s_n = (3/4)^n$

As n increases, the area keeps decreasing and **gets closer and closer to 0**.

? **The number of black triangles increases very quickly as the stages increase. Can you explain why?**

✔ ANSWER

At every stage the number is **multiplied by 3** (not just increased by a fixed amount), so it grows like 3^n: 1, 3, 9, 27, 81, 243, 729, ...

? **Can you explain why the area of the black region at Stage n will be $(3/4)^n$?**

✔ ANSWER

Going from one stage to the next, every black triangle is cut into 4 equal parts and 1 part is removed, so 3/4 of the black area remains.

Starting from 1 and multiplying by 3/4 for each of the n stages gives 1 × 3/4 × 3/4 × ... (n times) = **$(3/4)^n$**.

📝 Exercise Set 8.3

1 **Find the 12th term of a GP with common ratio 2, whose 8th term is 192.**

✔ ANSWER

$t_8 = ar^7 = 192 \rightarrow a \times 128 = 192 \rightarrow a = 3/2$

$t_{12} = ar^{11} = t_8 \times r^4 = 192 \times 16$

$t_{12} =$ **3072**

2 **Find the 10th and nth terms of the GP: 5, 25, 125, ...**

✔ ANSWER

$a = 5, r = 5$

$t_n = 5 \times 5^{n-1} =$ **5^n**

$t_{10} = 5^{10} =$ **9765625**

***3** **A sequence is given by $t_1 = 2$, $t_{n+1} = 3t_n - 2$ for $n \geq 1$. Which term of the sequence is 730?**

✔ **ANSWER**

$t_2 = 3 \times 2 - 2 = 4$

$t_3 = 3 \times 4 - 2 = 10$

$t_4 = 3 \times 10 - 2 = 28$

$t_5 = 3 \times 28 - 2 = 82$

$t_6 = 3 \times 82 - 2 = 244$

$t_7 = 3 \times 244 - 2 = 730$

730 is the **7th term**. (In fact $t_n = 3^{n-1} + 1$.)

4 **Which term of the GP: 2, 6, 18, ... is 4374? Write the explicit formula as well as the recursive formula for the nth term.**

✔ **ANSWER**

$a = 2$, $r = 3$, $t_n = 2 \times 3^{n-1}$

$2 \times 3^{n-1} = 4374 \rightarrow 3^{n-1} = 2187 = 3^7 \rightarrow n - 1 = 7$

$\rightarrow n = 8$

4374 is the **8th term**.

Explicit: $\mathbf{t_n = 2 \times 3^{n-1}}$

Recursive: $\mathbf{t_1 = 2, t_n = 3t_{n-1}}$ for $n \geq 2$

5 **A ball is dropped from a height of 80 m. Each time it bounces back to 60% of the height from which it fell.**

(i) What height does the ball reach after the 5th bounce?

(ii) What is the total vertical distance travelled by the time it hits the ground for the 6th time?

✔ ANSWER

Heights after bounces: 80 × 0.6 = 48, 28.8, 17.28, 10.368, 6.2208 (GP with r = 0.6)

(i) After 5th bounce: $80 \times 0.6^5 = 80 \times 0.07776$ = **6.2208 m** (about 6.22 m)

(ii) First fall = 80 m. Hits ground 1st time.

After each of the next 5 bounces it goes up and comes down the same height before the 2nd, 3rd, ..., 6th hit.

Up-heights total = 48 + 28.8 + 17.28 + 10.368 + 6.2208 = 110.6688 m

Total = 80 + 2 × 110.6688 = 80 + 221.3376 = **301.3376 m** (about 301.34 m)

6 **Which term of the sequence 2, 2√2, 4, ... is 128?**

✔ ANSWER

a = 2, r = 2√2 ÷ 2 = √2

$t_n = 2 \times (\sqrt{2})^{n-1} = 128 \rightarrow (\sqrt{2})^{n-1} = 64 = 2^6 = (\sqrt{2})^{12}$

n − 1 = 12 → n = 13

128 is the **13th term**.

7 **Sierpiński square carpet (Fig. 8.12): Stage 0 is a square; each square is divided into 9 equal**

squares and the centre one is removed, keeping 8.

(i) How many red squares are there in Stages 0 to 3?

(ii) Predict the number of red squares in Stages 4 and 5.

(iii) Find a rule (explicit and recursive) for the number of red squares at the nth stage.

(iv) If the area at Stage 0 is 1 square unit, find the red area in Stages 1 to 5, the explicit and recursive formula for Stage n, and what happens as n increases.

✔ ANSWER

Each red square becomes 8 smaller red squares at the next stage.

(i) Stage 0: **1**, Stage 1: **8**, Stage 2: **64**, Stage 3: **512**

(ii) Stage 4: 512 × 8 = **4096**, Stage 5: 4096 × 8 = **32768**

(iii) Explicit: $\mathbf{t_n = 8^n}$

Recursive: $\mathbf{t_0 = 1, t_n = 8 \times t_{n-1}}$ for n ≥ 1

(iv) Each stage keeps 8/9 of the red area.

Stage 1: **8/9**, Stage 2: **64/81**, Stage 3: **512/729**

Stage 4: **4096/6561**, Stage 5: **32768/59049** square unit

Explicit: $\mathbf{s_n = (8/9)^n}$; recursive: $\mathbf{s_0 = 1, s_n = (8/9) \times s_{n-1}}$ for n ≥ 1

As n increases, the red area keeps decreasing and **gets closer and closer to 0**.

End-of-Chapter Exercises

1 **Find the 31st term of an AP whose 11th term is 38 and 16th term is 73.**

✔ ANSWER

$a + 10d = 38$ ... (1)

$a + 15d = 73$ ... (2)

(2) − (1): $5d = 35 \rightarrow d = 7$

$a = 38 - 70 = -32$

$t_{31} = a + 30d = -32 + 210 =$ **178**

2 **Determine the AP whose third term is 16 and whose 7th term exceeds the 5th term by 12.**

✔ ANSWER

$t_7 - t_5 = (a + 6d) - (a + 4d) = 2d = 12 \rightarrow d = 6$

$a + 2d = 16 \rightarrow a = 16 - 12 = 4$

AP: **4, 10, 16, 22, 28, ...**

***3** **How many three-digit numbers are divisible by 7?**

✔ ANSWER

Smallest: 105 (7 × 15). Largest: 994 (7 × 142).

AP: 105, 112, ..., 994 with d = 7

$994 = 105 + (n - 1) \times 7 \rightarrow 889 = 7(n - 1) \rightarrow n - 1 = 127 \rightarrow n = 128$

128 numbers

***4** **How many multiples of 4 lie between 10 and 250?**

✔ ANSWER

Smallest: 12. Largest: 248.

$248 = 12 + (n - 1) \times 4 \rightarrow 236 = 4(n - 1) \rightarrow n - 1 = 59 \rightarrow n = 60$

60 multiples

***5** **Find a GP for which the sum of the first two terms is −4 and the fifth term is 4 times the third term.**

✔ ANSWER

$ar^4 = 4ar^2 \rightarrow r^2 = 4 \rightarrow r = 2$ or $r = -2$

$a + ar = -4 \rightarrow a(1 + r) = -4$

If $r = 2$: $3a = -4 \rightarrow a = -4/3$ → GP: **−4/3, −8/3, −16/3, −32/3, ...**

If $r = -2$: $-a = -4 \rightarrow a = 4$ → GP: **4, −8, 16, −32, ...**

Both GPs satisfy the conditions.

***6** **Find all possible ways of expressing 100 as the sum of consecutive natural numbers.**

✔ ANSWER

Let k consecutive numbers start at a: sum = $k(2a + k - 1)/2 = 100$, so $k(2a + k - 1) = 200$.

k and $(2a + k - 1)$ must be one odd and one even, with $2a + k - 1 > k$.

- k = 5: 2a + 4 = 40 → a = 18 → 18 + 19 + 20 + 21 + 22 = 100
- k = 8: 2a + 7 = 25 → a = 9 → 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 = 100

Other factor choices give a ≤ 0.

Two ways: 18 + 19 + 20 + 21 + 22 and 9 + 10 + ... + 16 (apart from 100 itself).

***7 The number of bacteria in a culture doubles every hour. If there were 30 bacteria originally, how many will be present at the end of the 2nd hour, 4th hour and nth hour?**

✔ ANSWER

After n hours: 30×2^n

End of 2nd hour: 30 × 4 = **120**

End of 4th hour: 30 × 16 = **480**

End of nth hour: **30×2^n**

***8 The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP.**

✔ ANSWER

(a + 3d) + (a + 7d) = 24 → 2a + 10d = 24 ... (1)

(a + 5d) + (a + 9d) = 44 → 2a + 14d = 44 ... (2)

(2) − (1): 4d = 20 → d = 5

2a = 24 − 50 = −26 → a = −13

First three terms: **−13, −8, −3**

***9** **Find the smallest value of n such that the sum of the first n natural numbers is greater than 1,000.**

✔ ANSWER

$n(n + 1)/2 > 1000 \rightarrow n(n + 1) > 2000$

$n = 44$: $44 \times 45 = 1980 \rightarrow$ sum 990 (not enough)

$n = 45$: $45 \times 46 = 2070 \rightarrow$ sum 1035 > 1000

n = 45

***10** **Which term of the GP: 2, 8, 32, ... is 131072? Write the explicit formula as well as the recursive formula for the nth term.**

✔ ANSWER

$a = 2$, $r = 4$, $t_n = 2 \times 4^{n-1}$

$2 \times 4^{n-1} = 131072 \rightarrow 4^{n-1} = 65536 = 4^8 \rightarrow n = 9$

131072 is the **9th term**.

Explicit: $\mathbf{t_n = 2 \times 4^{n-1}}$

Recursive: $\mathbf{t_1 = 2, t_n = 4t_{n-1}}$ for $n \geq 2$

***11** **The sum of the first three terms of a GP is 13/12 and their product is −1. Find the common ratio and the terms.**

✔ ANSWER

Take the terms as a/r, a, ar.

Product $= a^3 = -1 \rightarrow a = -1$

Sum: $-1/r - 1 - r = 13/12 \rightarrow r + 1/r = -25/12$

$12r^2 + 25r + 12 = 0 \rightarrow (4r + 3)(3r + 4) = 0$

$r = \mathbf{-3/4}$ or $r = \mathbf{-4/3}$

$r = -3/4$: terms $4/3, -1, 3/4$

$r = -4/3$: terms $3/4, -1, 4/3$

Terms: **4/3, −1, 3/4** (or in reverse order).

***12** **If the 4th, 10th and 16th terms of a GP are x, y and z respectively, prove that x, y, z are in GP.**

✔ ANSWER

1. Let the GP have first term a and common ratio r.

2. $x = ar^3$, $y = ar^9$, $z = ar^{15}$.

3. $y \div x = ar^9 \div ar^3 = r^6$.

4. $z \div y = ar^{15} \div ar^9 = r^6$.

5. $y/x = z/y$ (both equal r^6), so the ratio is constant.

Hence x, y, z are in GP (with common ratio r^6).

***13** **The sum of the first three terms of a GP is 26, and the sum of their squares is 364. Find the terms of the GP.**

✔ ANSWER

Terms a, ar, ar^2.

$a(1 + r + r^2) = 26$ ... (1)

$a^2(1 + r^2 + r^4) = 364$ ... (2)

Use $1 + r^2 + r^4 = (1 + r + r^2)(1 - r + r^2)$.

(2) ÷ (1): $a(1 - r + r^2) = 14$ ... (3)

(1) − (3): $2ar = 12 \rightarrow ar = 6$

(1) + (3): $2a(1 + r^2) = 40 \rightarrow a + ar^2 = 20$

Put $a = 6/r$: $6/r + 6r = 20 \rightarrow 3r^2 - 10r + 3 = 0 \rightarrow (3r - 1)(r - 3) = 0$

$r = 3 \rightarrow a = 2$; $r = 1/3 \rightarrow a = 18$

Terms: **2, 6, 18** (or 18, 6, 2).

Check: 2 + 6 + 18 = 26; 4 + 36 + 324 = 364 ✓

***14** **Suppose $P_1 = 1$, $P_2 = 2$ and for $n > 2$, $P_n = P_1 + P_2 + \ldots + P_{n-1} + 1$. Find $P_1, P_2, \ldots, P_8$. Can you find a simpler recursive formula for P_n? Can you give an explicit formula?**

✔ ANSWER

$P_3 = 1 + 2 + 1 = 4$

$P_4 = 1 + 2 + 4 + 1 = 8$

$P_5 = 1 + 2 + 4 + 8 + 1 = 16$

Similarly $P_6 = 32$, $P_7 = 64$, $P_8 = 128$

Values: **1, 2, 4, 8, 16, 32, 64, 128**

Since $P_1 + \ldots + P_{n-2} + 1 = P_{n-1}$, we get $P_n = P_{n-1} + P_{n-1}$.

Simpler recursive rule: $\mathbf{P_1 = 1, P_n = 2P_{n-1}}$ for $n \geq 2$

Explicit formula: $\mathbf{P_n = 2^{n-1}}$

***15** Suppose $W_1 = 1$, $W_2 = 2$ and for $n > 2$, $W_n = W_1 + W_2 + \ldots + W_{n-2} + 2$. Find $W_1, W_2, \ldots, W_8$. Do you recognise this sequence?

✔ ANSWER

$W_3 = W_1 + 2 = 3$

$W_4 = 1 + 2 + 2 = 5$

$W_5 = 1 + 2 + 3 + 2 = 8$

$W_6 = 1 + 2 + 3 + 5 + 2 = 13$

$W_7 = 1 + 2 + 3 + 5 + 8 + 2 = 21$

$W_8 = 1 + 2 + 3 + 5 + 8 + 13 + 2 = 34$

Values: **1, 2, 3, 5, 8, 13, 21, 34**

Each term is the sum of the previous two, so this is the **Virahānka–Fibonacci sequence**.

🎉 **Chapter 8 done! Revise the answers once more tomorrow.**